Some Problems

On Euclidean Geometry

Juan S. A. Siagian

Bandung, Indonesia · October 19, 2026

1. Preliminaries

1.1 Points and Vectors

We work in $\mathbb{R}^2$, whose elements are called points or vectors depending on their role. Below, $\boldsymbol{x}$, $\boldsymbol{y}$, $\boldsymbol{u}$ and v are **vectors** and $\boldsymbol{P}$ and $\boldsymbol{Q}$ are **points**.

Operations $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and $c\,(x_1, x_2) = (cx_1,\ cx_2)$.

Inner product $\langle x, y\rangle = x_1y_1 + x_2y_2$.

Length $|x| = \sqrt{\langle x, x\rangle}$.

Distance $d(P, Q) = |Q - P|$. With this distance, $\mathbb{R}^2$ is the Euclidean plane $\mathbb{E}^2$.

Parallel $u, v \neq 0$ and $v = tu$ for some $t \in \mathbb{R}$.

Same direction $u, v \neq 0$ and $v = tu$ for some $t > 0$.

Orthogonal $\langle u, v\rangle = 0$.

Perpendicular vector $v^{\perp} = (-v_2, v_1)$ for $v = (v_1, v_2)$.

Note.

1. Parallel and same direction differ only in the sign of t.
   Parallel vectors may point opposite ways.
   Vectors with the same direction may not.
2. The vector $v^{\perp}$ is a specific vector orthogonal to v.
   Built by turning v by 90° counterclockwise.

1.2 Lines

Below, $\boldsymbol{P}$, $\boldsymbol{Q}$ and $\boldsymbol{R}$ are **points** and $\boldsymbol{v}$ is a **vector**.

Span $[v] = \{tv \mid t \in \mathbb{R}\}$.

Line A set $P + [v] = \{P + tv \mid t \in \mathbb{R}\}$ with $v \neq 0$. Since $P + [0] = \{P\}$ is a single point.

Direction vector The vector v in $P + [v]$.

Between Q lies between P and R if $Q = P + s(R - P)$ for some $s \in [0, 1]$.

1.3 Maps

Below, $\boldsymbol{S}$ and $\boldsymbol{T}$ are **maps** from $\mathbb{E}^2$ to $\mathbb{E}^2$, $\boldsymbol{w}$ is a **vector** and $\boldsymbol{X}$ and $\boldsymbol{Y}$ are **points**.

Identity $\mathrm{id}(X) = X$.

Composition $(S \circ T)(X) = S(T(X))$, so T acts first.

Inverse T^{-1} is the map with $T \circ T^{-1} = T^{-1} \circ T = \mathrm{id}$, when it exists.

Translation $T_w(X) = X + w$.

Isometry T with $d(T(X), T(Y)) = d(X, Y)$ for all X and Y.

Note.

1. $T_a \circ T_b = T_{a+b}$ for all vectors a and b, and $T_0 = \mathrm{id}$. So $T_w^{-1} = T_{-w}$.
2. An isometry is injective, since $T(X) = T(Y)$ gives $d(X, Y) = 0$.
3. Reflections, rotations and glide reflections are defined in later sections.

1.4 Two Facts

We take for granted the Cauchy–Schwarz inequality $|\langle u, v\rangle| \leq |u||v|$.
With equality exactly when $u = 0$, $v = 0$, or u and v are parallel.

Fact 1 (Triangle inequality)**.** For all $u, v \in \mathbb{R}^2$ we have $|u + v| \leq |u| + |v|$. Equality holds if and only if $u = 0$, or $v = 0$, or u and v have the same direction.

Proof.
By Cauchy–Schwarz

$$|u + v|^2 = |u|^2 + 2\langle u, v\rangle + |v|^2 \leq |u|^2 + 2|u||v| + |v|^2 = (|u| + |v|)^2.$$

Equality holds exactly when $\langle u, v\rangle = |u||v|$, and this happens in the following cases.

- If $u = 0$ or $v = 0$, both sides are 0.
- If $v = tu$ with $t > 0$, then $\langle u, v\rangle = t|u|^2 = |u||v|$.
- Conversely, let $u, v \neq 0$ with $\langle u, v\rangle = |u||v|$.
  By Cauchy–Schwarz $v = tu$ with $t \neq 0$, and $t|u|^2 = |t||u|^2$ forces $t > 0$. □

Fact 2 (Re-anchoring a line)**.**

(a) If $X \in P + [v]$ and $c \neq 0$, then $P + [v] = X + [cv]$.
  In particular every line has a direction vector of length 1.

(b) If $P + [v] = P' + [v']$, then v' is a multiple of v.
  So any two direction vectors of a line are parallel.

Proof.

(a) Write $X = P + sv$. Then $P + tv = X + \frac{t-s}{c}(cv)$ and $X + r(cv) = P + (s + rc)v$.
  Which gives both inclusions. For the last claim take $X = P$ and $c = 1/|v|$.

(b) Both P' and $P' + v'$ lie in $P + [v]$, say $P' = P + sv$ and $P' + v' = P + rv$.
  Subtracting gives $v' = (r - s)v$. □

2. Lines

Problem 1 (The line through two points)
Let P and Q be distinct points.
(a) Show that $\ell = P + [Q - P]$ is a line containing P and Q.
(b) Show that every line containing P and Q is equal to ℓ.
Hence exactly one line contains P and Q. We call it the line PQ.

Solution.

(a) ℓ is a line because $Q - P \neq 0$, and it contains P and Q because

$$P = P + 0 \cdot (Q - P) \qquad \text{and} \qquad Q = P + 1 \cdot (Q - P).$$

(b) Let $\ell' = X + [v]$ be a line containing P and Q. Then

$$P = X + s_1 v \qquad \text{and} \qquad Q = X + s_2 v$$

for some $s_1, s_2 \in \mathbb{R}$. Subtracting gives $Q - P = (s_2 - s_1)v$, and $s_2 \neq s_1$ because $Q \neq P$. We prove both inclusions.

($\subseteq$) Let $Y \in \ell$, so $Y = P + t(Q - P)$ for some t.
Substituting $P = X + s_1 v$ and $Q - P = (s_2 - s_1)v$ gives

$$Y = X + s_1 v + t(s_2 - s_1)v = X + \big(s_1 + t(s_2 - s_1)\big)v \in \ell'.$$

($\supseteq$) Let $Y \in \ell'$, so $Y = X + rv$ for some r.
Substituting $X = P - s_1 v$ and $v = (Q - P)/(s_2 - s_1)$ gives

$$Y = P + (r - s_1)v = P + \frac{r - s_1}{s_2 - s_1}(Q - P) \in \ell.$$

Hence $\ell' = \ell$. □

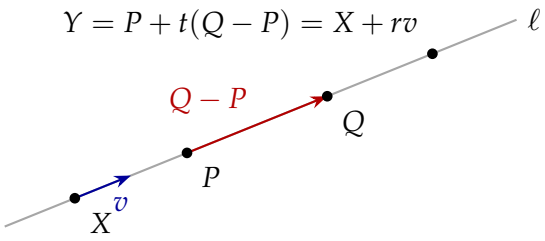


Figure 1. The line ℓ written as $P + [Q - P]$ and as $X + [v]$.

Problem 2 (Two lines meet at most once)
Show that two distinct lines have at most one point in common.

Solution.
Suppose two distinct lines ℓ_1 and ℓ_2 share two distinct points X and Y.
Applying Problem 1(b) to each line gives $\ell_1 = X + [Y - X] = \ell_2$.
This contradicts $\ell_1 \neq \ell_2$. □

Problem 3 (Betweenness and distance)
Recall that a point Q lies between P and R if $Q = P + s(R - P)$ for some $s \in [0, 1]$.
Show that for all points P, Q and R the equation

$$d(P, Q) + d(Q, R) = d(P, R)$$

holds if and only if Q lies between P and R.

Solution.
Put $u = R - Q$ and $v = Q - P$, so that $u + v = R - P$.
The equation then reads $|u| + |v| = |u + v|$.
($\Rightarrow$) By Fact 1 there are three cases.

- $v = 0$. Then $Q = P = P + 0 \cdot (R - P)$.
- $u = 0$. Then $Q = R = P + 1 \cdot (R - P)$.
- $u = tv$ with $t > 0$. Then $R - P = u + v = (1 + t)(Q - P)$, so

$$Q = P + s(R - P) \qquad \text{with} \qquad s = \frac{1}{1 + t}.$$

Since $t > 0$ we have $1 + t > 1$, so $0 < s < 1$.

In every case Q lies between P and R.

($\Leftarrow$) Let $Q = P + s(R - P)$ with $0 \leq s \leq 1$.
Then $Q - P = s(R - P)$ and $R - Q = (1 - s)(R - P)$, and since s, $1 - s \geq 0$

$$d(P, Q) + d(Q, R) = s|R - P| + (1 - s)|R - P| = |R - P| = d(P, R).$$ □

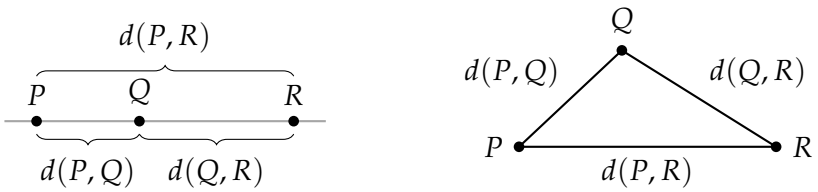


Figure 2. Left, Q lies between P and R. Right, Q is off the segment.

Problem 4 (Reading the parameter from distances)
Let $P \neq Q$ and let $X = (1-\lambda)P + \lambda Q$ for some $\lambda \in \mathbb{R}$.

(a) Show that
$$\lambda = \frac{d(P,X)^2 + d(P,Q)^2 - d(Q,X)^2}{2\,d(P,Q)^2}.$$

(b) Deduce that a point on PQ is determined by its distances to P and Q. That is, for points X and X' on the line PQ
$$d(P,X) = d(P,X') \text{ and } d(Q,X) = d(Q,X') \implies X = X'.$$

Solution.

(a) From $X = (1-\lambda)P + \lambda Q$ we get $X - P = \lambda(Q - P)$.
Writing $X - Q = (X - P) - (Q - P)$ and expanding the inner product gives
$$\begin{aligned} |X-Q|^2 &= |X-P|^2 - 2\langle X-P, Q-P\rangle + |Q-P|^2 \\ &= |X-P|^2 - 2\lambda|Q-P|^2 + |Q-P|^2. \end{aligned}$$
Since $P \neq Q$ we have $|Q-P|^2 \neq 0$, so we can solve for λ.
$$\lambda = \frac{|X-P|^2 + |Q-P|^2 - |X-Q|^2}{2|Q-P|^2} = \frac{d(P,X)^2 + d(P,Q)^2 - d(Q,X)^2}{2\,d(P,Q)^2}.$$

(b) By Problem 1 the line PQ is $P + [Q-P]$, so
$$X = (1-\lambda)P + \lambda Q \qquad \text{and} \qquad X' = (1-\lambda')P + \lambda' Q$$
for some $\lambda, \lambda' \in \mathbb{R}$.
By (a), λ and λ' come from the same formula in $d(P,\cdot)$, $d(Q,\cdot)$ and $d(P,Q)$.
These distances agree for X and X', so $\lambda = \lambda'$ and $X = X'$.
In Figure 3, each circle meets PQ twice and only X lies on both. □

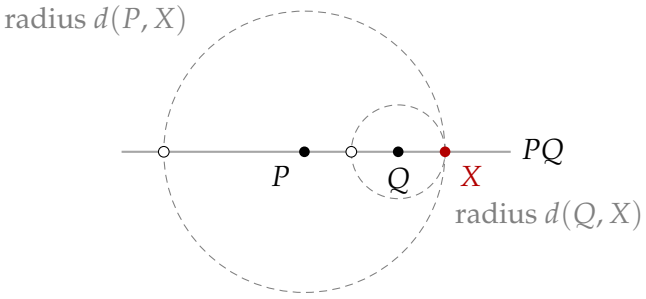


Figure 3. The circles about P and Q through X, and the line PQ.

Problem 5 (Orthonormal coordinates)
A pair $\{v, w\}$ of vectors is *orthonormal* if $|v| = |w| = 1$ and $\langle v, w\rangle = 0$.
Let $\{v, w\}$ be orthonormal.

(a) Show that v and w are linearly independent.

(b) Show that every $x \in \mathbb{R}^2$ satisfies $x = \langle x, v\rangle v + \langle x, w\rangle w$.

Solution.

(a) Suppose $av + bw = 0$ for some $a, b \in \mathbb{R}$.
Take the inner product with v and with w.
Since $\langle v, v\rangle = \langle w, w\rangle = 1$ and $\langle v, w\rangle = 0$ this gives

$$0 = \langle av + bw, v\rangle = a\langle v, v\rangle + b\langle w, v\rangle = a$$
$$0 = \langle av + bw, w\rangle = a\langle v, w\rangle + b\langle w, w\rangle = b.$$

Hence $a = b = 0$, so v and w are linearly independent.

(b) By (a), v and w are two independent vectors in the two-dimensional space $\mathbb{R}^2$.
So they form a basis, and $x = x_1 v + x_2 w$ for some $x_1, x_2 \in \mathbb{R}$.
Taking inner products as in (a) gives

$$\langle x, v\rangle = x_1\langle v, v\rangle + x_2\langle w, v\rangle = x_1$$
$$\langle x, w\rangle = x_1\langle v, w\rangle + x_2\langle w, w\rangle = x_2.$$

Hence $x = \langle x, v\rangle v + \langle x, w\rangle w$.
In Figure 4, these are the coordinates of x along the turned axes v and w. □

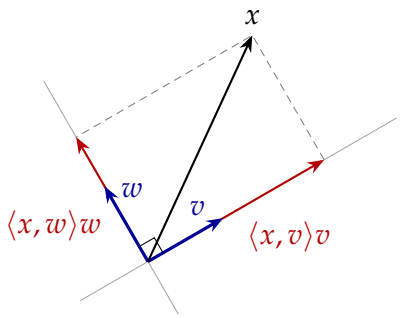


Figure 4. The vector x and its components along an orthonormal pair v, w.

Problem 6 (The perpendicular vector and the normal form of a line)
For $v = (v_1, v_2)$ let $v^\perp = (-v_2, v_1)$.

(a) Show that $\langle v, v^\perp \rangle = 0$ and $|v^\perp| = |v|$ for every v.
Deduce that $\{v, v^\perp\}$ is orthonormal whenever $|v| = 1$.

(b) Let P be a point and v a unit vector. Show that

$$P + [v] = \{X \in \mathbb{E}^2 \mid \langle X - P, v^\perp \rangle = 0\}.$$

Solution.

(a) A direct computation gives

$$\langle v, v^\perp \rangle = v_1(-v_2) + v_2 v_1 = 0$$
$$|v^\perp|^2 = (-v_2)^2 + v_1^2 = |v|^2.$$

If $|v| = 1$, then also $|v^\perp| = 1$, so $\{v, v^\perp\}$ is orthonormal.

(b) We prove both inclusions.

($\subseteq$) Let $X = P + tv$ for some t.
By (a), $\langle X - P, v^\perp \rangle = t\langle v, v^\perp \rangle = 0$.

($\supseteq$) Let $\langle X - P, v^\perp \rangle = 0$.
By (a), $\{v, v^\perp\}$ is orthonormal, so Problem 5 gives

$$X - P = \langle X - P, v \rangle v + \langle X - P, v^\perp \rangle v^\perp = \langle X - P, v \rangle v.$$

Hence $X = P + \langle X - P, v \rangle v \in P + [v]$.

In Figure 5, $X - P$ has no component along $v^\perp$, while $Y - P$ does. □

Remark. The unit length in (b) is only a convenience.
For any $v \neq 0$, Fact 2(a) gives $P + [v] = P + [v/|v|]$, and $(v/|v|)^\perp = v^\perp/|v|$.
So the same description holds.

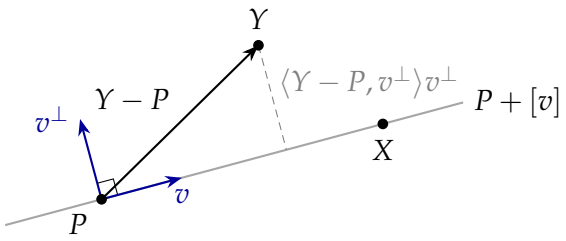


Figure 5. The line $P + [v]$ with a point X on it and a point Y off it.

Problem 7 (Parallel lines)
Two distinct lines are *parallel* if they have no point in common.
Let $\ell = P + [v]$ and $\ell' = P' + [v']$ be distinct lines.
Show that ℓ and ℓ' are parallel if and only if v' is a multiple of v.

Solution.

($\Leftarrow$) Let $v' = cv$, where $c \neq 0$ because $v' \neq 0$.
Suppose some point X lies on both lines. Then Fact 2(a) gives

$$\ell = X + [v] \qquad \text{and} \qquad \ell' = X + [v'] = X + [cv] = X + [v].$$

So $\ell = \ell'$, which is excluded. Hence $\ell \cap \ell' = \varnothing$.

($\Rightarrow$) We prove the contrapositive. Suppose v' is not a multiple of v.
Then v and v' are linearly independent.
Indeed, if $av + bv' = 0$ with $b \neq 0$, then $v' = -(a/b)v$, which is excluded.
So $b = 0$, and then $av = 0$ forces $a = 0$.
Hence $\{v, v'\}$ is a basis of $\mathbb{R}^2$, and $P' - P = tv - t'v'$ for some $t, t' \in \mathbb{R}$. The point

$$X = P + tv = P' + t'v'$$

lies on both lines, so they are not parallel. □

Remark. By Fact 2(b) this does not depend on the choice of direction vectors.
Together with Problem 2, two distinct lines fall into exactly one of two cases (Figure 6).

- Parallel direction vectors and no common point.
- Non-parallel direction vectors and exactly one common point.

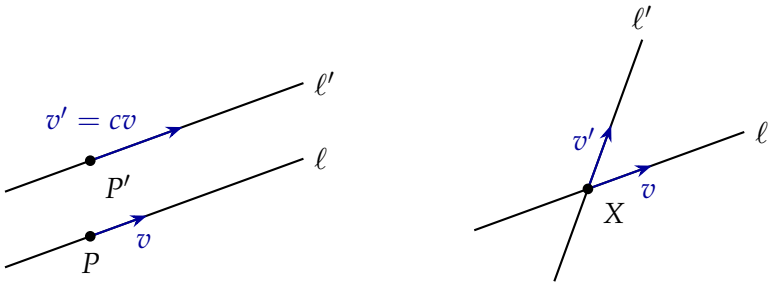


Figure 6. Left, two parallel lines. Right, two lines meeting at X.

3. Reflections

Below, $\ell = P + [v]$ is a **line** with $|v| = 1$, which is possible by Fact 2(a).
By Problem 6, $v^\perp$ is then a unit vector orthogonal to v.

Reflection The map $\Omega_\ell \colon \mathbb{E}^2 \to \mathbb{E}^2$ with $\Omega_\ell(X) = X - 2\langle X - P, v^\perp\rangle v^\perp$.

Note.

1. $\langle X - P, v^\perp\rangle v^\perp$ is the part of $X - P$ perpendicular to ℓ, and Ω_ℓ reverses it (Figure 7).
2. The formula uses a choice of P and v, but the map depends only on ℓ (Fact 3).

Fact 3 (The reflection depends only on the line)**.**
Let $\ell = P + [v] = P' + [v']$ with $|v| = |v'| = 1$.
Then for every point X

$$\langle X - P', v'^\perp\rangle v'^\perp = \langle X - P, v^\perp\rangle v^\perp.$$

So the two formulas for Ω_ℓ agree.

Proof.
By Fact 2(b), v' is a multiple of v, and both have length 1, so $v' = \pm v$.
Hence $v'^\perp = \pm v^\perp$, and the two signs cancel in $\langle X - P', v'^\perp\rangle v'^\perp$.
Also $P' \in \ell$, so $P' = P + sv$ for some s. Using $\langle v, v^\perp\rangle = 0$ we get

$$\langle X - P', v'^\perp\rangle v'^\perp = \langle X - P - sv, v^\perp\rangle v^\perp = \langle X - P, v^\perp\rangle v^\perp. \qquad \square$$

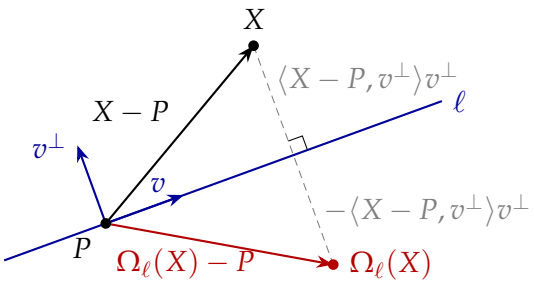


Figure 7. A point X and its reflection $\Omega_\ell(X)$ in the line ℓ.

Problem 8 (Reflections are bijections)
Let $\ell = P + [v]$ be a line with $|v| = 1$, and let Ω_ℓ be the reflection in ℓ

$$\Omega_\ell(X) = X - 2\langle X - P, v^\perp\rangle v^\perp.$$

(a) Show that $\Omega_\ell \circ \Omega_\ell = \mathrm{id}$.
(b) Deduce that Ω_ℓ is a bijection and $\Omega_\ell^{-1} = \Omega_\ell$.

Solution.
Write $n = v^\perp$, a unit vector.

(a) Fix a point X and put $c = \langle X - P, n\rangle$. Since $\langle n, n\rangle = 1$

$$\langle \Omega_\ell(X) - P, n\rangle = \langle X - P - 2cn, n\rangle = c - 2c = -c.$$

So the component along n flips sign, as in Figure 7.
Applying Ω_ℓ again flips it back

$$\Omega_\ell(\Omega_\ell(X)) = \Omega_\ell(X) - 2(-c)n = X - 2cn + 2cn = X.$$

In Figure 8 (right), reflecting X twice brings it back to X.

(b) We check the two halves of being a bijection.
- *Injective.* If $\Omega_\ell(X) = \Omega_\ell(Y)$, applying Ω_ℓ and using (a) gives $X = Y$.
- *Surjective.* Every point Y equals $\Omega_\ell(\Omega_\ell(Y))$.

Since $\Omega_\ell \circ \Omega_\ell = \mathrm{id}$, the inverse of Ω_ℓ is Ω_ℓ itself.
In Figure 8 (left), each point of K is paired with exactly one point of $\Omega_\ell(K)$. □

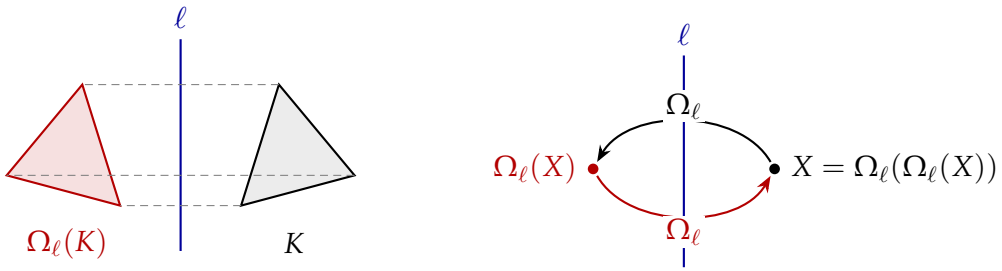


Figure 8. Left, a triangle K and its reflection. Right, a point X reflected twice.

Problem 9 (Reflections are isometries)

Let v be a unit vector.

(a) Show that every $w \in \mathbb{R}^2$ satisfies

$$w - 2\langle w, v^\perp\rangle v^\perp = \langle w, v\rangle v - \langle w, v^\perp\rangle v^\perp.$$

Deduce that $|w - 2\langle w, v^\perp\rangle v^\perp| = |w|$.

(b) Let $\ell = P + [v]$ and let Ω_ℓ be the reflection in ℓ.
Show that Ω_ℓ is an isometry, that is, for all points X_1 and X_2

$$d(\Omega_\ell(X_1), \Omega_\ell(X_2)) = d(X_1, X_2).$$

Solution.

(a) By Problem 6(a), $\{v, v^\perp\}$ is orthonormal, so Problem 5 gives

$$w = \langle w, v\rangle v + \langle w, v^\perp\rangle v^\perp.$$

Subtracting $2\langle w, v^\perp\rangle v^\perp$ from both sides gives the identity.
For the length, every $a, b \in \mathbb{R}$ satisfy

$$|av + bv^\perp|^2 = a^2\langle v, v\rangle + 2ab\langle v, v^\perp\rangle + b^2\langle v^\perp, v^\perp\rangle = a^2 + b^2.$$

Take $a = \langle w, v\rangle$ and $b = \pm\langle w, v^\perp\rangle$.
Then w and $w - 2\langle w, v^\perp\rangle v^\perp$ both have squared length $\langle w, v\rangle^2 + \langle w, v^\perp\rangle^2$.

(b) Subtracting the formulas for $\Omega_\ell(X_1)$ and $\Omega_\ell(X_2)$ gives

$$\begin{aligned}\Omega_\ell(X_1) - \Omega_\ell(X_2) &= (X_1 - X_2) - 2\big(\langle X_1 - P, v^\perp\rangle - \langle X_2 - P, v^\perp\rangle\big)v^\perp \\ &= (X_1 - X_2) - 2\langle X_1 - X_2, v^\perp\rangle v^\perp.\end{aligned}$$

By (a) with $w = X_1 - X_2$, this vector has length $|X_1 - X_2| = d(X_1, X_2)$.
In Figure 9, the segment and its image have the same length. □

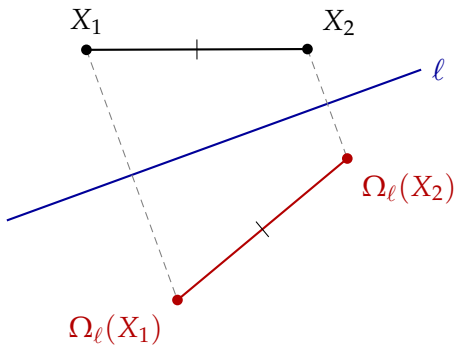


Figure 9. A segment X_1X_2 and its reflection in ℓ.

Problem 10 (The mirror is the fixed set)
Let ℓ be a line and Ω_ℓ the reflection in ℓ.
Show that $\Omega_\ell(X) = X$ if and only if $X \in \ell$.
In other words, the points that Ω_ℓ does not move are exactly the points of ℓ.

Solution.
Write $\ell = P + [v]$ with $|v| = 1$. Since $v^\perp \neq 0$

$$\Omega_\ell(X) = X \iff 2\langle X - P, v^\perp\rangle v^\perp = 0 \iff \langle X - P, v^\perp\rangle = 0 \iff X \in \ell.$$

The last step is Problem 6(b). □

Problem 11 (The mirror as an equidistant set)
Let ℓ be a line and Q a point not on ℓ.
Show that a point X lies on ℓ if and only if $d(X, Q) = d(X, \Omega_\ell(Q))$.

Solution.
Write $\ell = P + [v]$ with $|v| = 1$ and put $n = v^\perp$.

($\Rightarrow$) If $X \in \ell$, then $\Omega_\ell(X) = X$ by Problem 10.
Since Ω_ℓ is an isometry (Problem 9) we get

$$d(X, \Omega_\ell(Q)) = d(\Omega_\ell(X), \Omega_\ell(Q)) = d(X, Q).$$

($\Leftarrow$) Put $a = \langle X - P, n\rangle$, $q = \langle Q - P, n\rangle$ and $c = \langle X - P, v\rangle - \langle Q - P, v\rangle$.
By Problem 5 applied to $\{v, n\}$, and since $\Omega_\ell(Q) - P = (Q - P) - 2qn$

$$\begin{aligned} X - P &= \langle X - P, v\rangle v + an \\ Q - P &= \langle Q - P, v\rangle v + qn \\ \Omega_\ell(Q) - P &= \langle Q - P, v\rangle v - qn. \end{aligned}$$

Subtracting gives

$$X - Q = cv + (a - q)n \qquad \text{and} \qquad X - \Omega_\ell(Q) = cv + (a + q)n.$$

As in Problem 9(a), the squared distances are

$$d(X, Q)^2 = c^2 + (a - q)^2 \qquad \text{and} \qquad d(X, \Omega_\ell(Q))^2 = c^2 + (a + q)^2.$$

These are equal exactly when $(a + q)^2 - (a - q)^2 = 4aq = 0$.
Since $Q \notin \ell$, Problem 6(b) gives $q \neq 0$, so $a = 0$ and $X \in \ell$.

In Figure 10, X on ℓ is equally far from Q and $\Omega_\ell(Q)$, while Y off ℓ is not. □

Remark. The hypothesis $Q \notin \ell$ is needed only for ($\Leftarrow$).
If $Q \in \ell$, then $\Omega_\ell(Q) = Q$ and every point of the plane satisfies the equation.

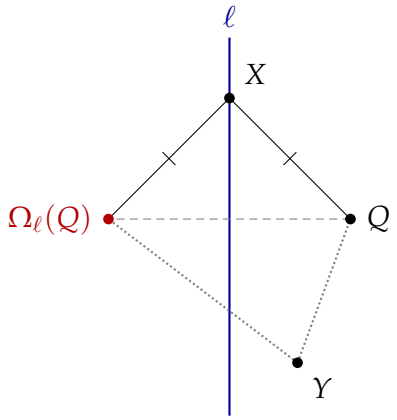


Figure 10. A point Q, its reflection, and points X on ℓ and Y off ℓ.

Problem 11 identifies the perpendicular bisector of a segment, which we use in Section 6.

Corollary 1 (Perpendicular bisector)**.**
Let $A \neq B$, and put $M = \frac{1}{2}(A + B)$ and $u = (B - A)/|B - A|$. Then

$$\{X \in \mathbb{E}^2 \mid d(X, A) = d(X, B)\} = M + [u^\perp].$$

Proof.
Let $\ell = M + [u^\perp]$. Its unit direction is $u^\perp$, and $(u^\perp)^\perp = -u$, so

$$\Omega_\ell(X) = X - 2\langle X - M, -u\rangle(-u) = X - 2\langle X - M, u\rangle u.$$

Since $B - M = \frac{1}{2}(B - A) = \frac{1}{2}|B - A|\, u$, we have $\langle B - M, u\rangle = \frac{1}{2}|B - A| \neq 0$.
Hence $B \notin \ell$ by Problem 6(b), and $\Omega_\ell(B) = B - |B - A|\, u = A$.
Problem 11 with $Q = B$ now shows that $X \in \ell$ exactly when $d(X, B) = d(X, A)$.
In Figure 11, Ω_ℓ swaps A and B, and X on ℓ is equally far from both. □

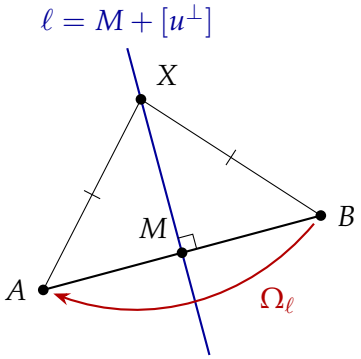


Figure 11. A segment AB, its midpoint M and its perpendicular bisector ℓ.

4. Rotations

Below, $\boldsymbol{\theta}$ is a **real number** and $\boldsymbol{P}$ is a **point**.
When a matrix acts on a point, the point is written as a column.

Rotation matrix $\mathrm{rot}(\theta) = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$, also used as the map $X \mapsto \mathrm{rot}(\theta)X$.

Rotation about $\boldsymbol{P}$ $R_{P,\theta}(X) = P + \mathrm{rot}(\theta)(X - P)$, that is, $R_{P,\theta} = T_P \circ \mathrm{rot}(\theta) \circ T_{-P}$.

Angle θ is an angle of the line $P + [(\cos\theta, \sin\theta)]$.

Perpendicular Two lines are perpendicular if their direction vectors are orthogonal.

Note.

1. T_P is the translation by the vector P.
   So $R_{P,\theta}$ moves P to the origin, turns by θ, and moves back.
2. Every unit vector is $(\cos\theta, \sin\theta)$ for some $\theta \in \mathbb{R}$, so every line has an angle.

Fact 4 (Reflections in lines through the origin)**.**
Let $v = (\cos\theta, \sin\theta)$. The reflection in $[v]$ is the matrix

$$S(2\theta) = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}.$$

Consequently, for $v_i = (\cos\theta_i, \sin\theta_i)$ with $i = 1, 2$

$$\Omega_{[v_2]} \circ \Omega_{[v_1]} = \mathrm{rot}\big(2(\theta_2 - \theta_1)\big).$$

Proof.
Put $n = v^\perp = (-\sin\theta, \cos\theta)$. Then $\Omega_{[v]}(x) = x - 2\langle x, n\rangle n = (I - 2nn^\top)x$, and

$$I - 2nn^\top = \begin{pmatrix} 1 - 2\sin^2\theta & 2\sin\theta\cos\theta \\ 2\sin\theta\cos\theta & 1 - 2\cos^2\theta \end{pmatrix} = S(2\theta).$$

For all a and b, the addition formulas give

$$S(a)S(b) = \begin{pmatrix} \cos a\cos b + \sin a\sin b & \cos a\sin b - \sin a\cos b \\ \sin a\cos b - \cos a\sin b & \sin a\sin b + \cos a\cos b \end{pmatrix} = \mathrm{rot}(a - b).$$

Taking $a = 2\theta_2$ and $b = 2\theta_1$ gives the second claim. □

Fact 5 (Moving the mirror off the origin)**.**
If $\ell = P + [v]$ with $|v| = 1$, then $\Omega_\ell = T_P \circ \Omega_{[v]} \circ T_{-P}$.

Proof.
For every point X we have $T_{-P}(X) = X - P$, so

$$T_P\big(\Omega_{[v]}(X - P)\big) = (X - P) - 2\langle X - P, v^\perp\rangle v^\perp + P = \Omega_\ell(X).$$ □

Problem 12 (Adding angles)
Let $\theta, \theta' \in \mathbb{R}$.

(a) Show that $\mathrm{rot}(\theta)\,\mathrm{rot}(\theta') = \mathrm{rot}(\theta+\theta')$.

(b) Deduce that $\mathrm{rot}(\theta)$ is invertible with inverse $\mathrm{rot}(-\theta)$.

(c) Deduce that $\mathrm{rot}(\theta)\,\mathrm{rot}(\theta') = \mathrm{rot}(\theta')\,\mathrm{rot}(\theta)$.

Solution.

(a) Multiplying out gives

$$\mathrm{rot}(\theta)\,\mathrm{rot}(\theta') = \begin{pmatrix} \cos\theta\cos\theta' - \sin\theta\sin\theta' & -(\sin\theta\cos\theta' + \cos\theta\sin\theta') \\ \sin\theta\cos\theta' + \cos\theta\sin\theta' & \cos\theta\cos\theta' - \sin\theta\sin\theta' \end{pmatrix}.$$

The addition formulas

$$\begin{aligned} \cos(\alpha+\beta) &= \cos\alpha\cos\beta - \sin\alpha\sin\beta \\ \sin(\alpha+\beta) &= \sin\alpha\cos\beta + \cos\alpha\sin\beta \end{aligned}$$

turn the entries into those of $\mathrm{rot}(\theta+\theta')$

$$\mathrm{rot}(\theta)\,\mathrm{rot}(\theta') = \begin{pmatrix} \cos(\theta+\theta') & -\sin(\theta+\theta') \\ \sin(\theta+\theta') & \cos(\theta+\theta') \end{pmatrix} = \mathrm{rot}(\theta+\theta').$$

In Figure 12, turning e_1 by θ' and then by θ lands at angle $\theta+\theta'$.

(b) By (a) with $\theta' = -\theta$, and since $\mathrm{rot}(0) = I$

$$\mathrm{rot}(\theta)\,\mathrm{rot}(-\theta) = \mathrm{rot}(0) = I = \mathrm{rot}(-\theta)\,\mathrm{rot}(\theta).$$

So $\mathrm{rot}(\theta)$ is invertible with inverse $\mathrm{rot}(-\theta)$.

(c) By (a), both products equal $\mathrm{rot}(\theta+\theta') = \mathrm{rot}(\theta'+\theta)$. □

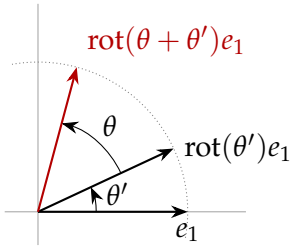


Figure 12. The vector e_1 turned by θ' and then by θ.

Problem 13 (Two mirrors through a point)
Let P be a point.
For $i = 1, 2$ let $\theta_i \in \mathbb{R}$, $v_i = (\cos\theta_i, \sin\theta_i)$ and $\ell_i = P + [v_i]$. Show that

$$\Omega_{\ell_2} \circ \Omega_{\ell_1} = T_P \circ \mathrm{rot}\big(2(\theta_2 - \theta_1)\big) \circ T_{-P} = R_{P,2(\theta_2-\theta_1)}.$$

In words, reflecting in two lines through P is the same as rotating about P by twice the angle from the first line to the second.

Solution.
By Fact 5, $\Omega_{\ell_i} = T_P \circ \Omega_{[v_i]} \circ T_{-P}$ for $i = 1, 2$. Since $T_{-P} \circ T_P = \mathrm{id}$

$$\begin{aligned}\Omega_{\ell_2} \circ \Omega_{\ell_1} &= T_P \circ \Omega_{[v_2]} \circ (T_{-P} \circ T_P) \circ \Omega_{[v_1]} \circ T_{-P} \\ &= T_P \circ \big(\Omega_{[v_2]} \circ \Omega_{[v_1]}\big) \circ T_{-P} \\ &= T_P \circ \mathrm{rot}\big(2(\theta_2 - \theta_1)\big) \circ T_{-P}.\end{aligned}$$

The last step is Fact 4.
In Figure 13, the mirrors meet at angle $\alpha = \theta_2 - \theta_1$ and X turns about P by 2α. □

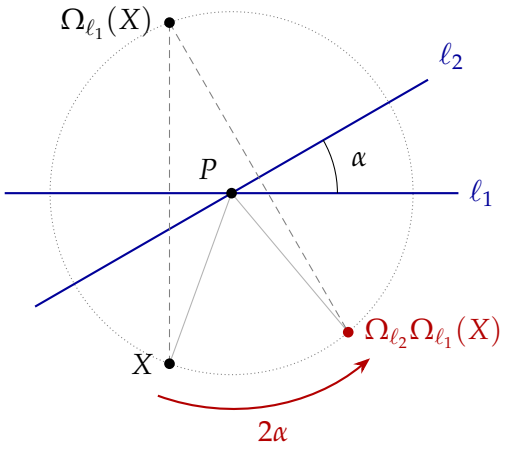


Figure 13. Two mirrors through P at angle α, and a point X reflected in both.

Remark. Three consequences are used below.

(i) *Only the angle matters.*
The composition depends on the two lines only through P and $\theta_2 - \theta_1$.
So pairs of lines through P with the same angle difference give the same rotation.

(ii) *Perpendicular mirrors give a half-turn.*
If $\theta_2 - \theta_1 = \pi/2 + k\pi$ with $k \in \mathbb{Z}$, then $\mathrm{rot}\big(2(\theta_2 - \theta_1)\big) = -I$.
So $\Omega_{\ell_2} \circ \Omega_{\ell_1} = R_{P,\pi}$, the map $X \mapsto 2P - X$.

(iii) *Equal lines.*
The lines ℓ_1 and ℓ_2 coincide exactly when $\theta_2 - \theta_1$ is a multiple of π.
Equal lines have $v_2 = \pm v_1$ by Fact 2(b), and $v_2 = \pm v_1$ gives equal lines by Fact 2(a).

Problem 14 (Three mirrors through a point)
Let P be a point, let v_1, v_2, v_3 be unit vectors and let $\ell_i = P + [v_i]$.
Show that there is a unit vector v_4 such that the line $\ell_4 = P + [v_4]$ satisfies

$$\Omega_{\ell_3} \circ \Omega_{\ell_2} \circ \Omega_{\ell_1} = \Omega_{\ell_4}.$$

Solution.
The idea is to trade the pair ℓ_2, ℓ_3 for a pair ℓ_1, ℓ_4 with the same angle difference.
Write $v_i = (\cos\theta_i, \sin\theta_i)$ for $i = 1, 2, 3$, and set

$$\theta_4 = \theta_1 - \theta_2 + \theta_3, \qquad v_4 = (\cos\theta_4, \sin\theta_4), \qquad \ell_4 = P + [v_4].$$

Since $\theta_4 - \theta_1 = \theta_3 - \theta_2$, Problem 13 gives

$$\Omega_{\ell_4} \circ \Omega_{\ell_1} = R_{P,2(\theta_4-\theta_1)} = R_{P,2(\theta_3-\theta_2)} = \Omega_{\ell_3} \circ \Omega_{\ell_2}.$$

Composing on the right with Ω_{ℓ_1} and using $\Omega_{\ell_1} \circ \Omega_{\ell_1} = \mathrm{id}$ (Problem 8) gives

$$\Omega_{\ell_4} = \Omega_{\ell_4} \circ \Omega_{\ell_1} \circ \Omega_{\ell_1} = \Omega_{\ell_3} \circ \Omega_{\ell_2} \circ \Omega_{\ell_1}.$$

In Figure 14, ℓ_4 is ℓ_1 turned about P by the angle $\beta = \theta_3 - \theta_2$ from ℓ_2 to ℓ_3. □

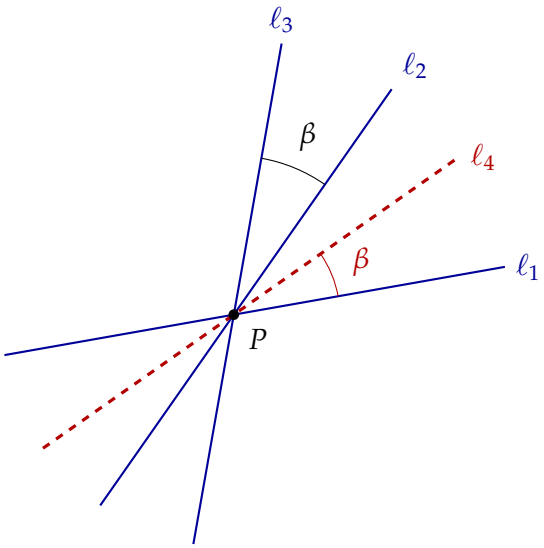


Figure 14. Three mirrors ℓ_1, ℓ_2, ℓ_3 through P and the fourth mirror ℓ_4.

5. Glide Reflections

Below, $\boldsymbol{\ell} = P + [v]$ is a **line** with $|v| = 1$ and $n = v^{\perp}$, and $\boldsymbol{w}$ and $\boldsymbol{y}$ are **vectors**.

Parallel to $\boldsymbol{\ell}$ w is a multiple of v.

Glide reflection The map $T_w \circ \Omega_\ell$ with $w \neq 0$ parallel to ℓ.

Axis The line ℓ of a glide reflection $T_w \circ \Omega_\ell$.

Vector reflection $S_\ell(y) = y - 2\langle y, n\rangle n$, which flips the component of y along n.

Note.

1. Unlike parallel vectors in 1.1, a vector parallel to ℓ may be 0.
2. As in Fact 3, S_ℓ does not depend on the sign of n.

Fact 6 (Reflections and translations)**.**
For every vector y we have $\Omega_\ell \circ T_y = T_{S_\ell(y)} \circ \Omega_\ell$.

Proof.
For every point X

$$\Omega_\ell(X + y) = X + y - 2\langle X - P, n\rangle n - 2\langle y, n\rangle n = \Omega_\ell(X) + S_\ell(y).$$

In words, Ω_ℓ turns a translation by y into a translation by $S_\ell(y)$. □

Fact 7 (Parallel mirrors)**.**
Let $\ell = A + [v]$ and $m = B + [v]$ with $|v| = 1$, and put $n = v^{\perp}$. Then

$$\Omega_m \circ \Omega_\ell = T_{2\langle B-A,n\rangle n}.$$

The vector $\langle B - A, n\rangle n$ is 0 exactly when $\ell = m$.

Proof.
Fix a point X and put $c = \langle X - A, n\rangle$.
As in Problem 8(a), $\langle \Omega_\ell(X) - A, n\rangle = -c$.
Hence $\langle \Omega_\ell(X) - B, n\rangle = -c - \langle B - A, n\rangle$, and therefore

$$\begin{aligned}\Omega_m(\Omega_\ell(X)) &= \Omega_\ell(X) + 2\big(c + \langle B - A, n\rangle\big)n \\ &= X - 2cn + 2cn + 2\langle B - A, n\rangle n \\ &= X + 2\langle B - A, n\rangle n.\end{aligned}$$

By Problem 6(b), $\langle B - A, n\rangle = 0$ exactly when $B \in \ell$.
By Fact 2(a) this happens exactly when $m = \ell$. □

Problem 15 (Sliding along the mirror)
Let $\ell = P + [v]$ be a line with $|v| = 1$, and let w be a vector parallel to ℓ.
Show that $T_w \circ \Omega_\ell = \Omega_\ell \circ T_w$.

Solution.
Write $w = cv$. Then $\langle w, v^\perp \rangle = c\langle v, v^\perp \rangle = 0$, so $S_\ell(w) = w$.
Fact 6 now gives $\Omega_\ell \circ T_w = T_{S_\ell(w)} \circ \Omega_\ell = T_w \circ \Omega_\ell$.
In Figure 15, both paths from X end at the same point. □

Remark. So a glide reflection can be written in either order.
Problem 17 shows that the condition on w cannot be dropped.

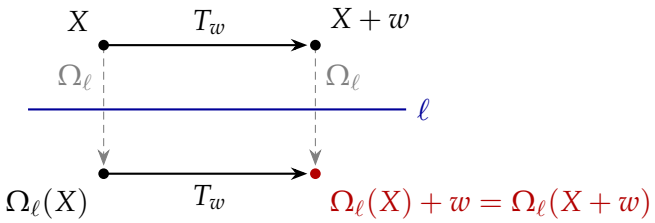


Figure 15. A point X reflected in ℓ and translated by w, in both orders.

Problem 16 (The square of a glide reflection)
Let $\gamma = T_w \circ \Omega_\ell$ be a glide reflection, so $w \neq 0$ is parallel to ℓ.
(a) Show that $\gamma \circ \gamma = T_{2w}$.
(b) Deduce that γ has no fixed point.

Solution.
(a) The middle factors commute by Problem 15, and $\Omega_\ell \circ \Omega_\ell = \mathrm{id}$ by Problem 8. So

$$\gamma \circ \gamma = T_w \circ \Omega_\ell \circ T_w \circ \Omega_\ell = T_w \circ T_w \circ \Omega_\ell \circ \Omega_\ell = T_{2w}.$$

(b) Suppose $\gamma(X) = X$. Then also $\gamma(\gamma(X)) = X$.
But by (a), $\gamma(\gamma(X)) = X + 2w \neq X$, since $w \neq 0$.
In Figure 16, no footprint stays put, and two steps make T_{2w}. □

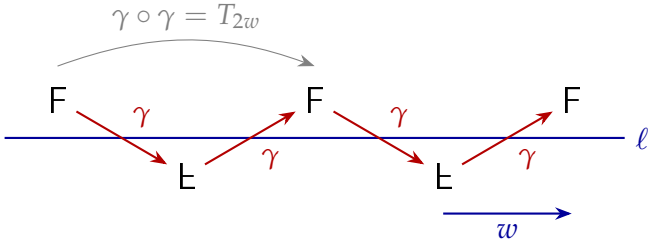


Figure 16. Footprints moved along the axis ℓ by a glide reflection γ.

Problem 17 (A commutator)
Let $\ell = P + [u]$ be a line with $|u| = 1$, put $n = u^\perp$, and let v be any vector.
The *commutator* of T_v and Ω_ℓ is

$$[T_v, \Omega_\ell] = T_v \circ \Omega_\ell \circ T_{-v} \circ \Omega_\ell.$$

(a) Show that $[T_v, \Omega_\ell] = T_{2\langle v,n\rangle n}$.
This is the translation by twice the component of v perpendicular to ℓ.
(b) Show that $[T_v, \Omega_\ell] = \mathrm{id}$ if and only if v is parallel to ℓ.
Show that this happens exactly when T_v and Ω_ℓ commute.

Solution.

(a) Fact 6 with $y = -v$ gives

$$\Omega_\ell \circ T_{-v} = T_{S_\ell(-v)} \circ \Omega_\ell = T_{-S_\ell(v)} \circ \Omega_\ell.$$

With $\Omega_\ell \circ \Omega_\ell = \mathrm{id}$ (Problem 8) this gives

$$[T_v, \Omega_\ell] = T_v \circ T_{-S_\ell(v)} \circ \Omega_\ell \circ \Omega_\ell = T_{v-S_\ell(v)} = T_{2\langle v,n\rangle n}.$$

In Figure 17, the part of v along ℓ cancels and the part along n doubles.

(b) A translation T_y is the identity exactly when $y = 0$.
So $[T_v, \Omega_\ell] = \mathrm{id}$ exactly when $\langle v, n\rangle = 0$.
By Problem 5 applied to $\{u, n\}$ we have $v = \langle v, u\rangle u + \langle v, n\rangle n$.
So $\langle v, n\rangle = 0$ exactly when v is a multiple of u, that is, parallel to ℓ.
Finally, since $T_{-v} = T_v^{-1}$ and $\Omega_\ell = \Omega_\ell^{-1}$

$$[T_v, \Omega_\ell] = \mathrm{id} \iff T_v \circ \Omega_\ell = (T_{-v} \circ \Omega_\ell)^{-1} = \Omega_\ell \circ T_v. \qquad \square$$

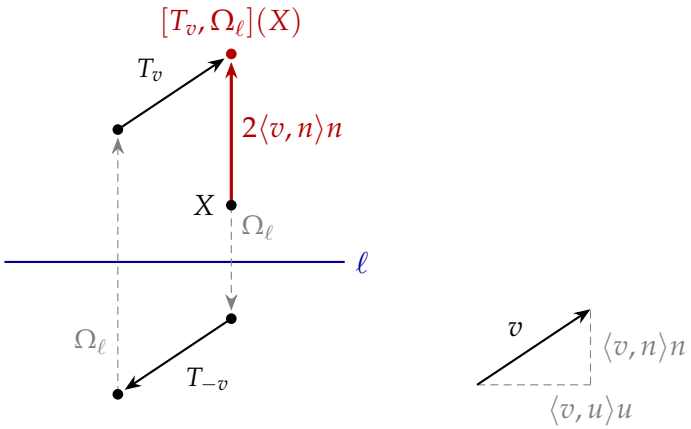


Figure 17. The four steps of $[T_v, \Omega_\ell]$ applied to X, and the two parts of v.

Problem 18 (Three mirrors in general position)
The following terms are for three lines, of which some may coincide.

Concurrent Some point lies on all three lines.

Share a direction Their direction vectors are pairwise parallel.

A glide reflection is a map $T_w \circ \Omega_k$ with k a line and $w \neq 0$ parallel to k.
Let ℓ_1, ℓ_2, ℓ_3 be lines that are neither concurrent nor share a direction.
Show that $\Omega_{\ell_3} \circ \Omega_{\ell_2} \circ \Omega_{\ell_1}$ is a glide reflection.

Solution.
Write $T = \Omega_{\ell_3} \circ \Omega_{\ell_2} \circ \Omega_{\ell_1}$.
The idea is to turn the mirrors without changing T until two are parallel.
Recall that lines with non-parallel directions meet exactly once (Problems 2 and 7).

Case 1. The direction vectors of ℓ_1 and ℓ_2 are not parallel.
Then ℓ_1 and ℓ_2 meet at a single point A, and $A \notin \ell_3$ because the lines are not concurrent.
By Fact 2 we can write

$$\ell_1 = A + [(\cos\alpha_1, \sin\alpha_1)], \qquad \ell_2 = A + [(\cos\alpha_2, \sin\alpha_2)], \qquad \ell_3 = C + [(\cos\beta, \sin\beta)]$$

for some point C and angles α_1, α_2 and β.

Step 1. Turn the pair ℓ_1, ℓ_2 about A (Figure 18(b)).
Let ℓ_2' be the line through A with angle $\beta + \pi/2$. It is perpendicular to ℓ_3.
Let ℓ_1' be the line through A with angle $\alpha_1' = \alpha_1 + (\beta + \pi/2) - \alpha_2$.
The angle from ℓ_1' to ℓ_2' is again $\alpha_2 - \alpha_1$, so Problem 13 gives

$$\Omega_{\ell_2} \circ \Omega_{\ell_1} = R_{A,2(\alpha_2-\alpha_1)} = \Omega_{\ell_2'} \circ \Omega_{\ell_1'}.$$

As $\ell_1 \neq \ell_2$ and the angle differences agree, $\ell_1' \neq \ell_2'$ by Remark (iii) after Problem 13.
Hence $T = \Omega_{\ell_3} \circ \Omega_{\ell_2'} \circ \Omega_{\ell_1'}$.
Step 2. Turn the pair ℓ_2', ℓ_3 about B (Figure 18(c)).
The lines ℓ_2' and ℓ_3 have orthogonal direction vectors, so they meet at a single point B.
Since $A \in \ell_2'$ but $A \notin \ell_3$, we have $B \neq A$.
Both lines pass through B, so Remark (ii) after Problem 13 gives $\Omega_{\ell_3} \circ \Omega_{\ell_2'} = R_{B,\pi}$.
Let $u = (\cos\alpha_1', \sin\alpha_1')$, a unit direction of ℓ_1', and put $m = B + [u]$ and $k = B + [u^\perp]$.
These are perpendicular lines through B, so again $R_{B,\pi} = \Omega_k \circ \Omega_m$. Hence

$$T = \Omega_k \circ \Omega_m \circ \Omega_{\ell_1'}.$$

Step 3. The parallel pair ℓ_1', m gives a translation.
The lines $\ell_1' = A + [u]$ and $m = B + [u]$ share the direction u, so Fact 7 gives

$$\Omega_m \circ \Omega_{\ell_1'} = T_w \qquad \text{with} \qquad w = 2\langle B - A, u^\perp \rangle u^\perp.$$

Suppose $w = 0$. Then Fact 7 gives $m = \ell_1'$, so $B \in \ell_1'$.
But $B \in \ell_2'$, and the distinct lines ℓ_1' and ℓ_2' meet only at A (Problem 2).
So $B = A$, which is false. Hence $w \neq 0$.

Step 4. Conclusion (Figure 18(d)).
Now $T = \Omega_k \circ T_w$, where $w \neq 0$ is a multiple of $u^\perp$, the direction of k.
By Problem 15, $T = T_w \circ \Omega_k$, a glide reflection with axis k.
In Figure 18(d), the three reflections (dotted) and the glide send X to the same point.

Case 2. The direction vectors of ℓ_1 and ℓ_2 are parallel.
Then those of ℓ_2 and ℓ_3 are not, since otherwise all three lines would share a direction.
By Problem 8, $T^{-1} = \Omega_{\ell_1} \circ \Omega_{\ell_2} \circ \Omega_{\ell_3}$.
This is the same kind of product for the lines ℓ_3, ℓ_2, ℓ_1 taken in this order.
These lines satisfy the hypotheses, and the first two have non-parallel direction vectors.
By Case 1, $T^{-1} = T_w \circ \Omega_k$ for some line k and some $w \neq 0$ parallel to k.
Then Problem 15 gives

$$T = \Omega_k^{-1} \circ T_w^{-1} = \Omega_k \circ T_{-w} = T_{-w} \circ \Omega_k.$$

This is a glide reflection, since $-w \neq 0$ is parallel to k. □

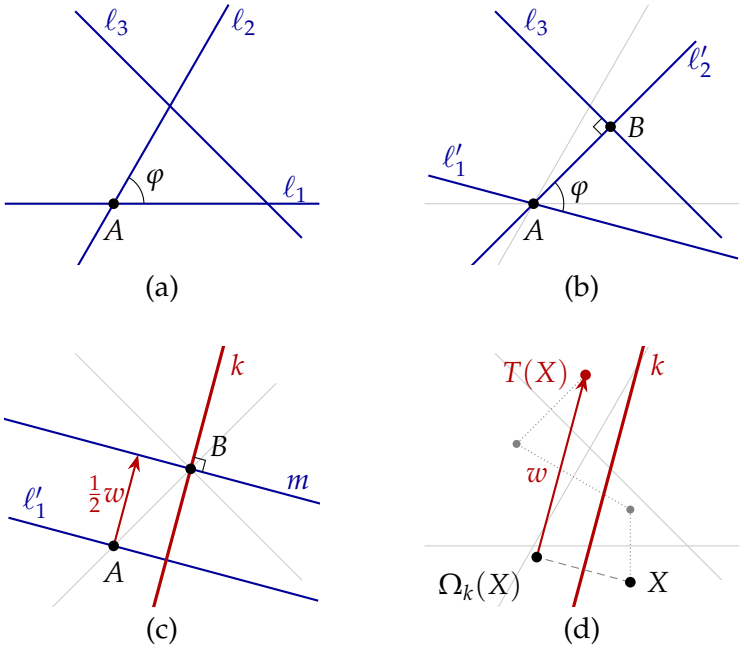


Figure 18. (a) The three mirrors. (b) After Step 1. (c) After Steps 2 and 3. (d) The glide reflection.

Problem 19 (When three reflections collapse to one)
Let ℓ_1, ℓ_2, ℓ_3 be lines, with concurrent and sharing a direction as in Problem 18.
Show that $\Omega_{\ell_3} \circ \Omega_{\ell_2} \circ \Omega_{\ell_1}$ is a reflection Ω_{ℓ_4} for some line ℓ_4
if and only if the three lines are concurrent or share a direction.

Solution.
Write $T = \Omega_{\ell_3} \circ \Omega_{\ell_2} \circ \Omega_{\ell_1}$.

($\Leftarrow$) There are two cases.

- *Concurrent.* Let P lie on all three lines.
  By Fact 2(a) we can write $\ell_i = P + [v_i]$ with unit vectors v_i.
  Problem 14 then gives a line ℓ_4 through P with $T = \Omega_{\ell_4}$.
- *Common direction.* By Fact 2(a) we can write $\ell_i = A_i + [v]$ with one unit v.
  Put $n = v^\perp$, $A_4 = A_1 + \langle A_3 - A_2, n\rangle n$ and $\ell_4 = A_4 + [v]$.
  Since $\langle A_4 - A_1, n\rangle = \langle A_3 - A_2, n\rangle$, Fact 7 applied to both pairs gives

$$\Omega_{\ell_4} \circ \Omega_{\ell_1} = T_{2\langle A_3 - A_2, n\rangle n} = \Omega_{\ell_3} \circ \Omega_{\ell_2}.$$

  Composing on the right with Ω_{ℓ_1} and using Problem 8 gives $\Omega_{\ell_4} = T$.
  So ℓ_4 is ℓ_1 shifted by the gap from ℓ_2 to ℓ_3, the parallel analogue of Problem 14.

($\Rightarrow$) Suppose the lines are neither concurrent nor share a direction.
By Problem 18, T is a glide reflection.
So T has no fixed point by Problem 16(b).
But every reflection Ω_ℓ fixes the points of ℓ (Problem 10).
Hence T is not a reflection. □

Remark. So a composition of three reflections is always a reflection or a glide reflection. The position of the mirrors tells which (Figure 19).

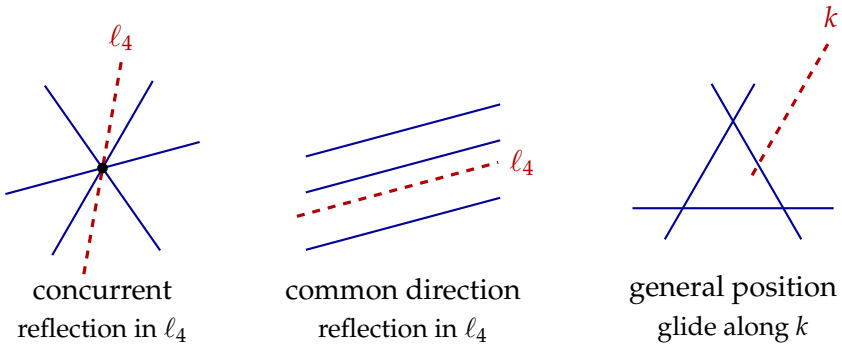


Figure 19. The three cases, with the resulting mirror ℓ_4 or glide axis k in red.

6. Isometries and Lines

Problem 20 (Isometries carry lines onto lines)
Let $T\colon \mathbb{E}^2 \to \mathbb{E}^2$ be an isometry, so $d(T(X), T(Y)) = d(X, Y)$ for all points X and Y.
Let ℓ be a line.

(a) Show that $T(\ell)$ is contained in a line.

(b) Show that $T(\ell)$ is a line.
More precisely, if $A \neq B$ are points of ℓ, show that for every $s \in \mathbb{R}$

$$T\big(A + s(B - A)\big) = T(A) + s\big(T(B) - T(A)\big).$$

Solution.

(a) Write $\ell = P + [v]$ with $|v| = 1$ and let $Q = P + v^{\perp}$.
Since $\langle Q - P, v^{\perp}\rangle = 1 \neq 0$, we have $Q \notin \ell$ by Problem 6(b).
Put $Q' = \Omega_\ell(Q)$. Then $Q' \neq Q$ by Problem 10, and Problem 11 gives

$$\ell = \{X \in \mathbb{E}^2 \mid d(X, Q) = d(X, Q')\}.$$

For $X \in \ell$, since T preserves distances

$$d(T(X), T(Q)) = d(X, Q) = d(X, Q') = d(T(X), T(Q')).$$

As T is injective, $T(Q) \neq T(Q')$. So by Corollary 1, $T(\ell)$ lies in the line

$$m = \{Y \in \mathbb{E}^2 \mid d(Y, T(Q)) = d(Y, T(Q'))\}.$$

In Figure 20, T keeps the equal distances from X to Q and to Q'.

(b) Let $A \neq B$ be points of ℓ, and put $A' = T(A)$ and $B' = T(B)$.
Then $A' \neq B'$ because T is injective, and both lie on m by (a).
By Problem 1, $\ell = A + [B - A]$ and $m = A' + [B' - A']$.
Put $\delta = |B - A|$. Then $\delta > 0$, and $\delta = |B' - A'|$ since T is an isometry.
Take $X = A + s(B - A)$ in ℓ. By (a), $T(X) = A' + \lambda(B' - A')$ for some $\lambda \in \mathbb{R}$.
Comparing distances to A' and to B' gives

$$|\lambda|\delta = d(T(X), A') = d(X, A) = |s|\delta$$
$$|\lambda - 1|\delta = d(T(X), B') = d(X, B) = |s - 1|\delta.$$

Hence $\lambda^2 = s^2$ and $(\lambda - 1)^2 = (s - 1)^2$.
Subtracting gives $-2\lambda + 1 = -2s + 1$, so $\lambda = s$, as Problem 4 predicts. Therefore

$$T\big(A + s(B - A)\big) = A' + s(B' - A') \qquad \text{for all } s \in \mathbb{R}.$$

As s runs over $\mathbb{R}$, the two sides run over $T(\ell)$ and m, so $T(\ell) = m$. □

Remark. With $s \in [0, 1]$, T maps points between A and B onto points between A' and B'.

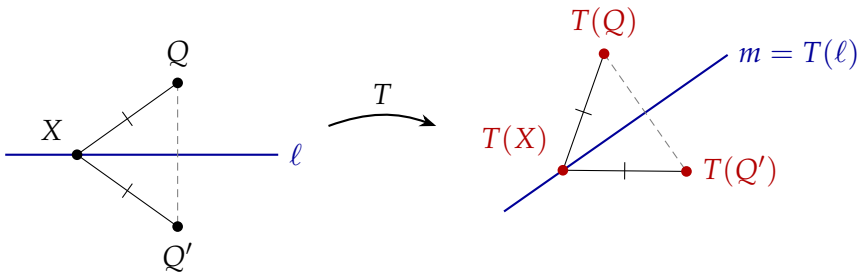


Figure 20. A line ℓ with points X, Q and $Q' = \Omega_\ell(Q)$, and their images under T.